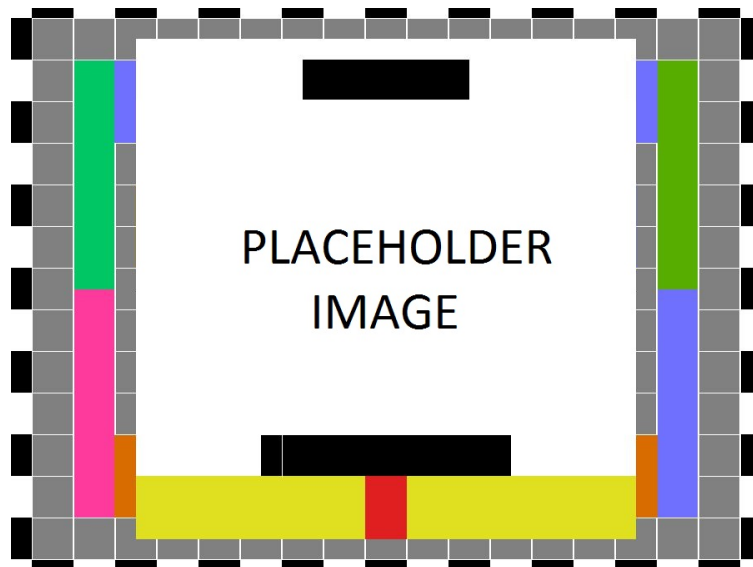


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**ARCHITECTURAL DESIGN SPECIFICATION
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM FULLSTACK
CIVICPULSE COMMUNITY SERVICE-REQUEST
DASHBOARD**

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REVISION HISTORY

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0.1	11.02.2026	JVN	document creation
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1 INTRODUCTION

Architecture of CivicPulse Community Service-Request Dashboard as an ISO/IEC/IEEE 42010:2022 [1] description using IEEE 1016-2009 [2] viewpoints. Refines the SRS. Scope: the web frontend, backend API, and datastore, plus SSO/geocoding/SMTP integrations.

2 STAKEHOLDERS & ARCHITECTURAL CONCERNS

Stakeholder	Primary concerns
City (sponsor)	workflow fit, security, privacy, availability, cost
Residents	easy submission, accessibility, status visibility
Staff agents	efficient triage/assignment, role-appropriate access
City IT	SSO integration, observability, deployability

3 ARCHITECTURE VIEWPOINTS

Context (Section 4), Composition/Data Flow (Section 5), Logical/Functional (Sections 6–8), Deployment (Section 8). Style: a classic three-tier web application — SPA presentation, stateless API, relational + geospatial datastore — deployed as containers behind a reverse proxy. Interfaces #NN; subsystems [Layer] . [Name].

4 SYSTEM OVERVIEW & CONTEXT VIEW

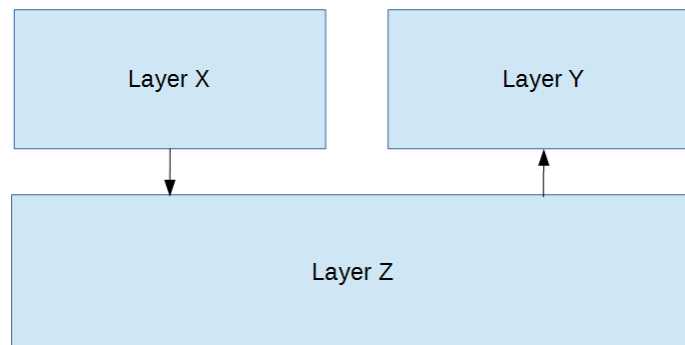


Figure 1: Three-tier web architecture

External actors: residents and staff (browsers), the city SSO (OIDC), the mapping API, and the SMTP relay. X renders UI; Y enforces logic, auth, and integrations; Z persists data.

5 SUBSYSTEM DEFINITIONS & DATA FLOW

6 X LAYER SUBSYSTEMS — PRESENTATION

6.1 X.RESIDENTPORTAL

Responsibilities: public SPA for submit/track (FR-01, FR-02), map picker (EI-02), accessible/responsive (US-01).

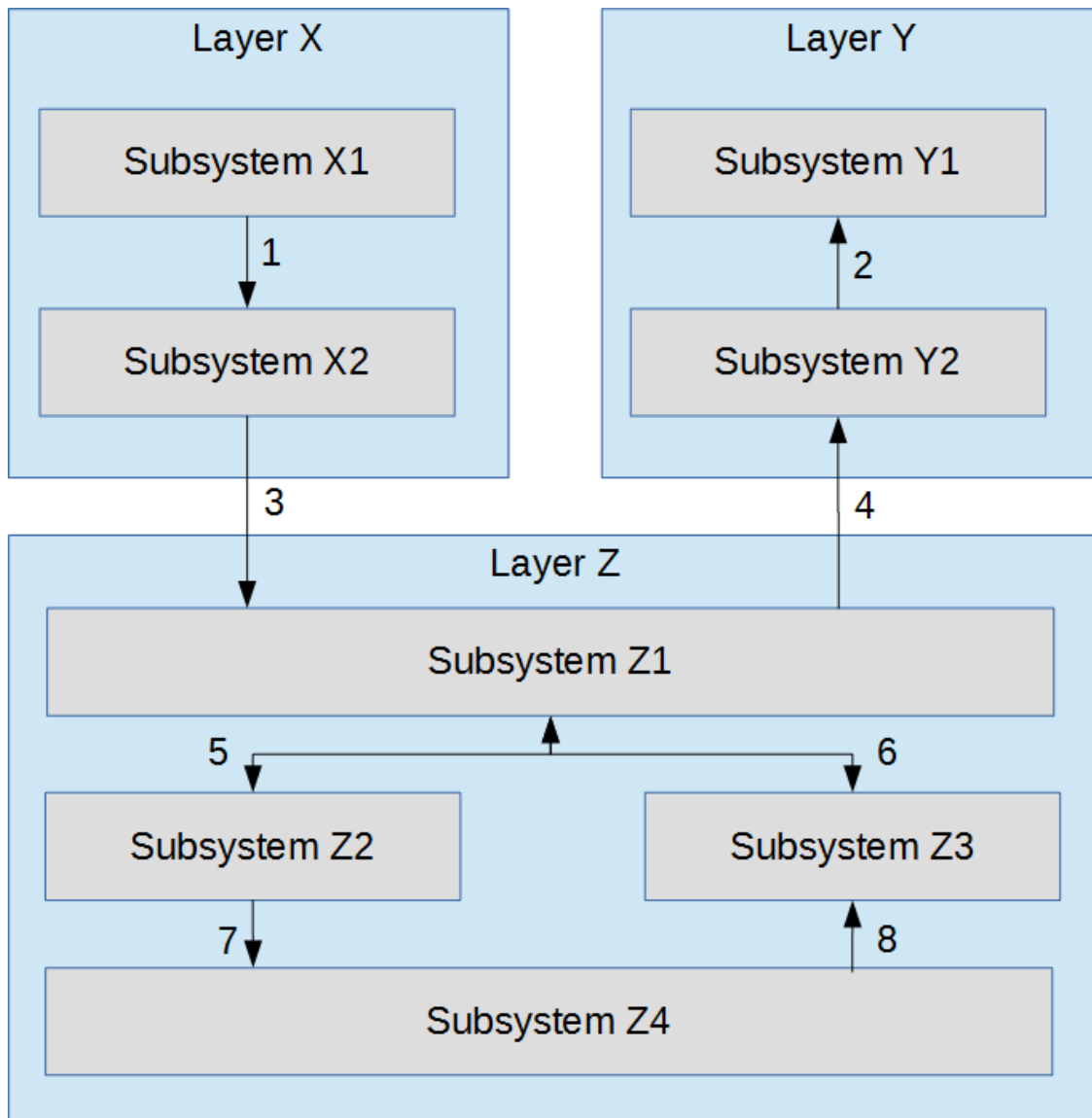


Figure 2: Browser → API → services → datastore/integrations

ID	Description	Inputs	Outputs
#01	Public REST calls	form data	request id, status

6.2 X.STAFFDASHBOARD

Responsibilities: authenticated SPA for triage/assignment/status (FR-03, FR-04); RBAC-aware UI (SE-01).

7 Y LAYER SUBSYSTEMS — APPLICATION

7.1 Y.APIGATEWAY

Responsibilities: single REST entry, TLS termination, OIDC validation, RBAC enforcement (SE-01).

ID	Description	Inputs	Outputs
#02	Authenticated API	requests + token	authorized calls

7.2 Y.REQUESTSERVICE

Responsibilities: request lifecycle, triage, assignment, status history (FR-01–FR-04); duplicate detection (FR-05).

7.3 Y.NOTIFICATIONSERVICE

Responsibilities: send status emails via SMTP on status change (FR-04).

8 Z LAYER SUBSYSTEMS — DATA

8.1 Z.DATASTORE

Responsibilities: PostgreSQL + PostGIS for requests, users, audit, and geospatial queries (FR-05 radius). **Deployment:** managed DB in the cloud project.

8.2 Z.OBJECTSTORE

Responsibilities: store request photos.

8.3 Z.INTEGRATIONS

Responsibilities: adapters for the geocoding API and SMTP relay, isolating external dependencies behind stable interfaces.

9 ARCHITECTURE DECISIONS & RATIONALE

9.1 AD-01: THREE-TIER WITH STATELESS API

Context: multi-role web workflow, need for horizontal scale (PE-01, QA-01). **Decision:** stateless API behind a proxy with a shared datastore. **Alternatives:** server-rendered monolith (harder to scale/mobile). **Consequences:** clean SPA/API split; requires token-based auth. **Affects:** PE-01, SE-01, QA-01.

9.2 AD-02: DELEGATED IDENTITY VIA OIDC

Context: EI-01/SE-01 — staff use city credentials. **Decision:** delegate to the city IdP; store no staff passwords. **Alternatives:** local accounts (more risk, poor fit). **Consequences:** depends on SSO availability. **Affects:** EI-01, SE-01.

10 REQUIREMENTS TRACEABILITY

Req ID	Requirement (abbrev.)	Realizing subsystem(s)
FR-01/02	Submit/track	X.ResidentPortal, Y.RequestService
FR-03	Triage/assign	X.StaffDashboard, Y.RequestService
FR-04	Status + notify	Y.RequestService, Y.NotificationService
FR-05	Duplicate detection	Y.RequestService, Z.Datastore
SE-01	RBAC + TLS	Y.APIGateway
EI-01	City SSO	Y.APIGateway

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 42010:2022 — Software, systems and enterprise — Architecture description,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2022.
- [2] IEEE, “IEEE Std 1016-2009 — IEEE Standard for Information Technology — Systems Design — Software Design Descriptions,” Institute of Electrical and Electronics Engineers, New York, NY, USA, Tech. Rep., 2009.